# Design and Implementation of a Dual-Evaporator Transcritical High Temperature Heat Pump

Research Project

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# 1 Context

According to the IEA report from March 2025, the global $CO_2$ emissions have increased by 0.8% in 2024, reaching an all-time high of 37.8 Gt $CO_2$ a year. In addition, the report highlights that the energy demand grew by 2.2% in 2024. This increase in $CO_2$ emissions and energy demand is totally inconsistent with the goals of the Paris Agreement, which aims to abate the $CO_2$ emissions by 43% by 2030 and to reach net-zero emissions by 2050.

In [2], we can read that the waste heat potential in the EU-27 and UK reached 221.32 TWh/year in 2021. In comparision, the demand for less than 140°C heat in Germany was 612 PJ, or 170 TWh in 2012 [1]. These numbers justify the interest from companies/researchers to design and study machines capable of recovering this waste energy.

In this work, we will focus on the use of high temperature heat pumps (HTHP's) for recovering waste heat from industrial processes. In [1], the authors provide a list of manufacturers and universities that have already been working on the design of several machines for a at least one decade. The already existing machines cover a large spectrum of applications, going from small machines of a few thousands watts to big ones of more than one megawatt. However, new design ideas can still be studied in order to improve the current recovery systems.

The main purpose of this work is to design the thermodynamical cycle of a transcritical dual-evaporator HTHP, build it and test its performances in laboratory conditions. No similar applications were found in the litterature. Beside, the machine build in the frame of this master thesis should be used in the future for an academic purpose.

# 2 Literature review

As discussed in the previous section, no article has been found on dual-evaporator transcritical HTHP's. However, a short liretature review on the current state of the art technology for HTHP's has been conducted and lead to the identification of several reasons why the topic is of particular interest.

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# 3 Research question

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# 4 Project description

In order to answer to the previous research question, we distinguish six steps in this master thesis. These steps can be observed in the Figure ...

## Preliminary cycle design

In this first part, we will numerically modeled the thermodynamical design. Each student will work simultaneously on different blocks, determined in advance. The analysis of the cycle will include an energy and exergy analysis to ensure optimal performances. The numerical model will be writen in Python and the module Coolprop will be used to access numerical thermodynamics tables.

## Final cycle design

Assembling all the different blocks will be the purpose of the second part. Beside connecting the blocks, an optimization process will be developped to determine the optimal parameters in nominal conditions. In addition, more physical aspects will be added to the model such as head losses computations and heat transfer modeling.

## Control

One main advantage of this machine will be its capacity of simulating different conditions thanks to the dual evaporator. A control logic must therefore be implemeted. This part is critical because we'll have to think about the nature and the location of the sensors. A student in industrial engineering will bring his expertise and will start to collaborate with us.

## Machine design

In this part, the other student will be in charge of the design of the machine. It will contain material/components selection, heat exchangers and insulation design and the elaboration of precise technical draw. That will be our task to correct the final cycle design (head losses, heat exchange, ...). We will also collaborate with Frank, the technician that will be responsible of the buiding of the phase. We surely think that its expertise will be needed for many design uncertainties.

## Building

The building phase will be ensured by Frank and other technicians of the CREDEM.

## Testing

Finally, the last part will be dedicated to the evaluation of the machine. Several tests will take place to ensure the good operation of the machine and to collect data in order to compare them to theoritical values. We don't think it will possible to have a correction phase in case of underperforming.

# 5 Share of workload and responsibilities

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The workload of this Master's thesis project will be shared equally. To maximize productivity, we plan to work in parallel on different subtasks. This productivity will be crucial since, as will be discussed in the next section, the sizing phase of the different components should be completed before the end of the first semester to be able to have the components on time for the prototyping phase to start early in the second semester. Nonetheless, we will collaborate closely on strategic subtasks, important design decisions and provide mutual support when required.

Weekly peer meetings will be organized throught the whole duration of the project. These meetings will be used to discuss the progress of the project, share knowledge, and make important decisions. In addition, we will meet our mentors on a regular basis (ideally weekly) to present important results, get feedback and discuss the main challenges. During the

During the prototyping phase, meetings with the workshop technician will be held on a regular basis too.

Additionally, our progress will be presented in a more formal way during monthly meetings with our supervisor.

# 6 Roadmap and planning

To be completed by HW (2 pages).

# References

[1] Cordin Arpagaus, Frédéric Bless, Michael Uhlmann, Jürg Schiffmann, and Stefan S Bertsch. High temperature heat pumps: Market overview, state of the art, research status, refrigerants, and application potentials. *Energy*, 152:985–1010, 2018.

[2] George Kosmadakis. Industrial waste heat potential and heat exploitation solutions. *Applied Thermal Engineering*, 246:122957, 2024.